Finite-length and vertical effects in nonlinear-kicker injection into a Korea Light Source design lattice

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Abstract

A nonlinear kicker magnet (NKM) injects off axis while cancelling the ideal centred stored-beam centroid kick, but the injected bunch samples a strongly nonlinear field. We study a 0.525 m four-loop wire NKM for the 4 GeV Korea Light Source (KLS), whose on-momentum dynamic aperture at the kicker is about 12 mm, with an exact three-dimensional filament field, a transfer line tracked from the booster and native nonlinear ring tracking. Trajectory displacement and the physical vertical field change the injection physics. The beam's own deflection inside the magnet shifts the matched angle by 0.2 mrad and makes capture tolerant to one sign of angle error, and the vertical field off the kick extremum defocuses the off-axis bunch, costing up to five percentage points of capture. At the selected operating point the NKM captures 93.0 % of the injected particles over 500 turns, compared with 97.0 % for an equally strong uniform kick. For the aligned kicker the ideal centroid kick vanishes by symmetry, while finite-beam moments change; transparency is set by the kicker-to-orbit offset (0.10 of the rms size at 0.2 mm). Under the assumed combined errors the mean capture is 44.4 % with ideal static orbit correction of the transfer line (40.5 % without); reducing the booster jitter to a quarter of its assumed value raises it to 70.0 %, and the remaining loss comes from the combined static errors.

Keywords: Beam dynamics; Accelerator modelling and simulations (multi-particle dynamics, single-particle dynamics); Instrumentation for particle accelerators and storage rings - high energy (linear accelerators, synchrotrons)

1 Introduction

Fourth-generation storage rings reduce the electron emittance by an order of magnitude or more with multibend-achromat lattices [1, 2]. The strong focusing and sextupoles that achieve this leave a small dynamic aperture (DA), which becomes the main constraint on injection. These rings also run in top-up mode [3]: the beam is replenished every few minutes during user operation, so each injection should leave the stored beam undisturbed. Where the DA is too small for off-axis injection, the beam can be injected on axis by swapping bunches with an accumulator [4] or injected into the longitudinal acceptance [5]. Where off-axis injection remains possible, the conventional four-kicker orbit bump leaves residual oscillations because the kicker pulses cannot be matched perfectly.

A single pulsed multipole whose field vanishes on the stored-beam axis avoids this. Injection with a pulsed quadrupole was demonstrated at the KEK PF-AR [6] and with a pulsed sextupole at the KEK Photon Factory [7]. A non-linear kicker (NLK) with an off-axis field maximum was developed at BESSY II [8, 9], and a multipole injection kicker (MIK) met a transparency specification of 10 % of the rms beam size at MAX IV and reached 2 % at SOLEIL [10, 11, 12];

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because commissioning pulsed-multipole injection is not straightforward, a single dipole kicker was added at MAX IV for the commissioning stage [13]. At SIRIUS, NLK injection was predicted to reach 99 % efficiency for a −9 mm DA but delivered about 85 ± 8 % with the design sextupoles, and online sextupole optimisation was needed [14, 15]; BESSY II likewise reached up to 97 % only with optimised sextupoles [9].

The price of transparency is a kick that varies across the injected bunch. Liu *et al.* [14] identified this nonlinearity and the injected-beam size as the main efficiency limit, and placing the bunch near the kick extremum is the natural remedy. Design studies usually describe the kicker by a kick map, the straight-line field integral applied at the magnet centre [16]. In a strongly nonlinear kicker this approximation is not controlled: the injected beam is deflected by several milliradians inside the magnet, and the vertical kick off the midplane follows from the field's curl-free structure, which requires a consistently signed vector map. Integrating the field along the actual path, as is done for insertion devices and fringe fields [17], resolves trajectory displacement while preserving the physical vector field.

The Korea Light Source (KLS) is a 4 GeV, 28-cell hybrid multibend-achromat ring of about 800 m circumference with a natural emittance of 62 pm [18]. This article studies these effects for a 0.525 m four-loop wire NKM in a KLS design lattice, whose on-momentum horizontal DA at the injection point extends from −12 to +13 mm. The kicker is modelled by the exact three-dimensional field of its current filaments, validated against the RADIA [19] kick map, and tracked through in slices. The contributions are: (i) the identification of the beam's self-deflection inside the magnet as the finite-length mechanism, and a magnet-length trade-off; (ii) the vertical defocusing of the off-axis bunch, previously hidden by a sign error in the tabulated vertical kick; (iii) transparency as an exact symmetry, with alignment and residual-field tolerances; and (iv) capture tolerance curves against booster jitter, kicker offset and injected emittance. The study is simulation-only and uses an ideal lattice with selected sensitivity scenarios to ring errors and chamber apertures; it makes no operational claim.

2 System and model

2.1 Storage ring

The supplied file `K4GSR_HBIv4-1` is identified by the author as the lattice used by Kim *et al.* at IPAC'25 [18]. It is a design version, not the latest KLS lattice; its file identifier is retained because an independent release date was not supplied. Later lattice versions may change the nonlinear acceptance, so results here apply to this version.

The KLS design lattice (file `K4GSR_HBIv4-1`; source circumference 799.30 m, 3481 elements) is tracked with the Accelerator Toolbox [20, 21] in six dimensions with RF and radiation (table 1). The ring is re-ordered to start at the NKM centre, which is the kick and observation point. The synchrotron period is about 290 turns and the horizontal damping time about 3950 turns. Losses arise at the septum blade and from dynamical instability; chamber apertures and ring errors enter only in the sensitivity studies of section 5. Figure 1 shows the DA from native tracking over 500 turns: on momentum, particles with $x' = 0$ are stable between −12 and +13 mm.

2.2 Kicker field and tracking

The 0.525 m four-loop magnet is a preliminary design for this study by Dong-Eon Kim (Pohang Accelerator Laboratory), as stated by the author; it is not the KLS baseline injection kicker. The baseline uses a four-kicker orbit bump and a 1 mm thin septum [22]. The alternative NKM scheme studied here retains an assumed 2 mm blade and a stored-beam edge at −16 mm; these are study geometry, not the published baseline septum. Loop powering has not been specified, so series and independent-loop cases are distinguished in the transparency tolerances.

Table 1: Principal parameters. Booster-beam parameters are design assumptions.

Ring energy, circumference	4 GeV, 799.30 m
Tunes, chromaticities	68.18/23.26, +6.0/ + 3.4
Emittance, energy spread	61.6 pm rad, 1.26×10^{-3}
Optics at NKM centre	$\beta_x = 16.27$ m, $\alpha_x = -0.145$, $\beta_y = 5.60$ m, $D_x = 0.3$ mm
Stored-beam size at NKM	$\sigma_x = 31.7\,\mu$m; $\varepsilon_y = 0.1\,\varepsilon_x$ assumed
Booster beam at extraction	$\beta_{x,y} = 7.56/12.27$ m, $\varepsilon_{x,y} = 100/10$ nm rad, $\sigma_\delta = 1.1 \times 10^{-3}$
Transfer line (BTS)	21.79 m, matched optics (peak $\beta_{x,y} = 31.4/50.7$ m)
Septum	stored-beam edge −16 mm, blade 2 mm, 2.26 m upstream of NKM centre
NKM	four loops, wires at $(\pm 5.30, \pm 5.30)$ and $(\pm 10.25, \pm 10.25)$ mm, length 525 mm, 3000 A; peak kick 4.89 mrad at $x = -8.6$ mm ($S = 0.85$)

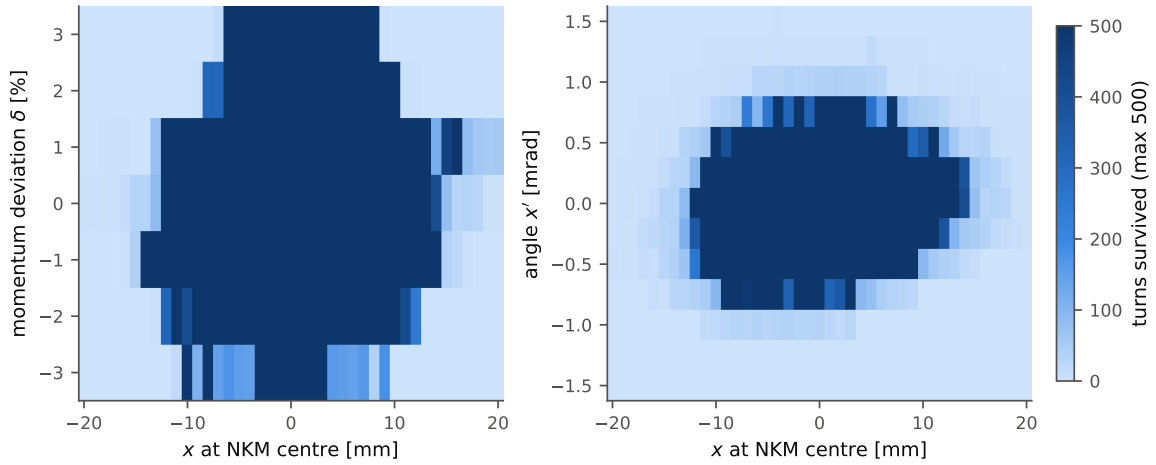


Figure 1: Dynamic aperture at the NKM centre from native tracking of single particles for 500 turns (1 mm grid in x): turns survived versus x and momentum deviation at $x' = 0$ (left) and versus x and x' at $\delta = 0$ (right).

The NKM consists of four closed rectangular filament loops, one per quadrant, each with an inner and an outer longitudinal wire joined by end connectors (figure 2, left). The field of straight filaments in vacuum is evaluated in closed form (Biot–Savart), including the fringe and end-connector fields. Its straight-line integrals reproduce both components of the RADIA kick map to 7×10^{-12} rad (4×10^{-10} relative) on 3 657 nodes, so the RADIA calculation and the filament model describe the same field. The tabulated map stores $(\int B_y \, \mathrm{d}s, \int B_x \, \mathrm{d}s)/B\rho$; the curl-free kick pair, $\partial \Delta x'/\partial y = \partial \Delta y'/\partial x$, requires $-\int B_x \, \mathrm{d}s$ for the vertical kick. Using the map's vertical sign therefore turns vertical defocusing into focusing; all thin-kick results below use the corrected sign.

The magnet is tracked in 640 drift–kick–drift slices over ± 0.45 m in canonical AT coordinates, $\mathrm{d}p_x/\mathrm{d}s = PS(B_y - y'B_s)/B\rho$ and $\mathrm{d}p_y/\mathrm{d}s = PS(-B_x + x'B_s)/B\rho$ with slopes $x' = p_x/(1+\delta)$, polarity P and field scale S. The kick acts on canonical momentum and carries no $1/(1+\delta)$ factor; the exit angle changes by 0.12 μrad per 10^{-3} in δ. Slice and span refinement change the exit angle by less than 7×10^{-11} rad. Particles within 1 mm of a conductor are counted as lost; none was. The polarity is chosen so that the kick at $x < 0$ cancels the angle of a beam travelling from the septum towards the axis (figure 2, right).

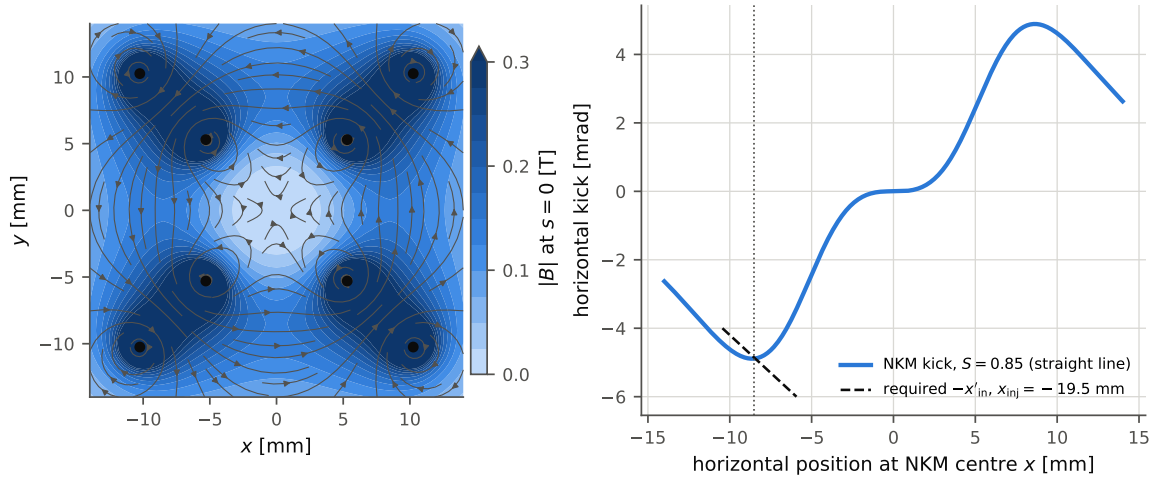


Figure 2: Left: NKM cross-section at $s = 0$ with the wire positions, field lines and $|B|$ of the exact filament field. Right: straight-line horizontal kick at $y = 0$ for $S = 0.85$ and the kick required to cancel the incoming angle of a beam launched from the septum at $x_{\mathrm{inj}} = -19.5$ mm; their intersection (dotted) is the thin-kick matched point.

An independent adaptive DOP853 integration checks the stated paraxial equations at the nominal point, the final interpolated angular edges (offsets -0.517/0.241 mrad), nonzero vertical position and angle, and $\delta = \pm 0.002$. Relative/absolute tolerances are $2 \times 10^{-11}/2 \times 10^{-13}$. Maximum 640-slice discrepancies are 3.26×10^{-11} m in x, 5.08×10^{-10} rad in p_x, 2.90×10^{-11} m in y and 5.06×10^{-10} rad in p_y. Refinement through 1280 slices and ± 0.9 m spans is below one percent of a 0.1σ response in either plane. The field is exact for the filament geometry; tracking is a validated paraxial approximation.

2.3 Injected beam, capture and controls

The injected bunch is generated at booster extraction and tracked element by element through the matched BTS line to the septum exit, where it is placed at $(x_{\mathrm{inj}}, x'_{\mathrm{inj}})$. It drifts to the magnet, is tracked through the 3-D NKM once (single-turn pulse, section 8) and then for 500 turns. *Capture* is the fraction of injected particles alive after the last turn; intervals are 95 % Wilson intervals [23] for particles and cluster-bootstrap intervals for error realizations. Lost particles are classified as lost at septum creation, inside the magnet, at the septum on a later

turn, or elsewhere in the ring; the categories reconcile with the particle count. The *matched* angle is the inner root of zero centroid angle after the magnet. Two controls receive identical particles: a uniform *dipole* kick and a *linearised* kick (with its curl-free vertical partner), both calibrated to the 3-D response of the nominal centroid.

3 Finite-length and vertical effects

At the operating point of section 4 the thin kick map gives a matched angle of 4.870 mrad, the 3-D magnet 4.6475 mrad. To isolate the cause, the map's horizontal profile was distributed over the magnet length and sampled either along the undeflected straight line or along the self-consistent deflected path (figure 3). Straight-path sampling gives 4.762 mrad; the deflected path gives 4.6476 mrad, equal to the 3-D result, and reproduces its response to angle errors to 0.01 mm. The finite-length effect is therefore the beam's own deflection inside the magnet, with a sagitta $|K|L/8 = 0.32$ mm, not path averaging or field details.

The effect is asymmetric. A positive septum angle error moves the trajectory inwards; on the straight line it would cross the kick extremum onto the weaker inner flank, leaving the extra angle under-compensated. The self-deflection instead holds the trajectory on the outer side of the extremum, where the kick is largest (figure 3, centre). A +0.3 mrad error changes the betatron amplitude of the kicked centroid by +1.22 mm with the thin kick but by -0.12 mm with the 3-D magnet, whereas −0.3 mrad gives +0.66 and +0.60 mm; the nominal amplitude is larger (9.57 against 9.07 mm). At fixed integrated kick, a longer magnet lowers the sensitivity to positive errors but raises the nominal amplitude, and for $L \geq 0.75$ m no matched solution remains (figure 3, right). A kick map is adequate only when the sagitta is small compared with the width of the kick extremum.

The vertical field matters as much. Near the kick extremum the 3-D magnet defocuses vertically: a centroid at $y = 1$ mm receives 0.25 mrad, against 0.13 mrad from the sign-corrected thin map, because the deflected path samples the field off the straight line. On identical bunches at the earlier operating point, capture is 91.1 % with the map's original vertical sign, 89.7 % with the corrected sign and 81.8 % with the 3-D magnet. Suppressing only the 3-D vertical kick restores 87.8 % (the horizontal-only finite-length model gives 87.4 %), so at that point the vertical defocusing costs about five percentage points. The linear vertical defocusing equals $-\partial\Delta x'/\partial x$ and vanishes at the kick extremum; at the operating point selected in section 4, where the centroid sits at the extremum, the cost is about two points (section 5).

4 Operating window

At the final point and reference chamber, the native ring captures 97.0 % at 500 turns, versus 100.0 % for an affine element-map reference on identical particles. The reference retains element lengths and explicit aperture checks at physical boundaries, and linearizes about the six-dimensional closed orbit with RF and radiation. A small-perturbation one-turn check agrees within 1.36×10^{-14} in the corresponding metre/radian/dimensionless coordinates. It deliberately omits nonlinear dynamics; all capture conclusions use native tracking.

Nominal native survival at 4000 turns is 96.1 % (95 % particle Wilson interval 94.7–97.1 %), a loss of 0.9 percentage points after turn 500. The 500-turn prefix reproduces the paired chamber-study particle masks exactly. Fresh bunches at the interpolated lower and upper angular-edge candidates give 79.5/85.0 % at 500 turns and 79.4/84.7 % at 4000 turns (95 % intervals 76.8–81.8 % and 82.3–86.8 %). The degraded corrected ring 5501 gives 3.0 % at 500 turns and 1.4 % at 4000 turns (0.8–2.3 %). Thus the stated window remains a 500-turn conditional contour, not an infinite-time acceptance or a guaranteed 4000-turn 80 % contour.

The point $x_{\mathrm{inj}} = -19.5$ mm, $S = 0.85$, $x'_{\mathrm{inj}} = 4.853$ mrad was selected by the largest particle-Wilson lower bound in a fixed matched-point grid using a 100/10 nm rad sensitivity beam. Four

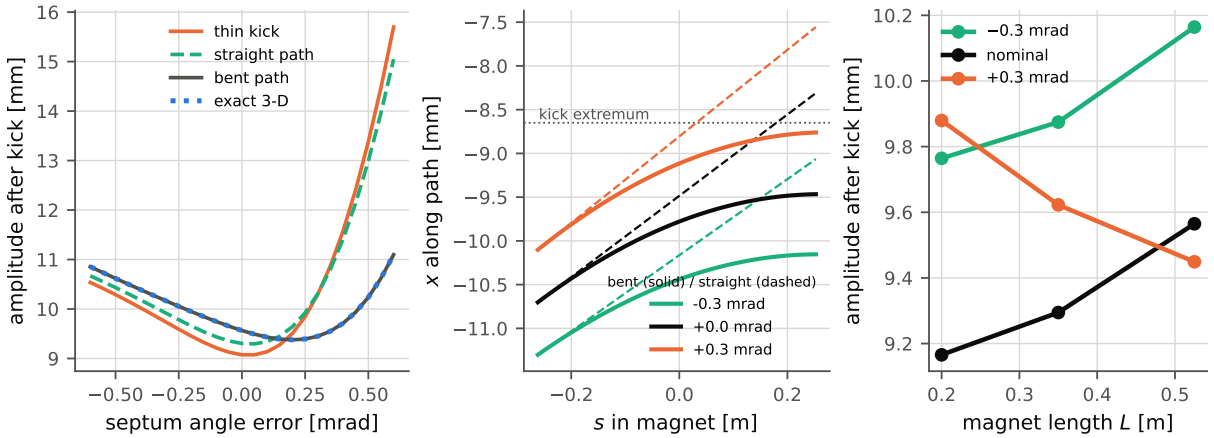


Figure 3: Finite-length mechanism. Left: betatron amplitude of the kicked centroid at the NKM centre versus septum angle error for the thin kick, the profile sampled on the straight and on the deflected path, and the exact 3-D field, each at its own matched angle. Centre: centroid path inside the magnet for angle errors of −0.3, 0 and +0.3 mrad (solid: deflected; dashed: straight) relative to the kick extremum. Right: deflected-path model versus magnet length at fixed integrated kick.

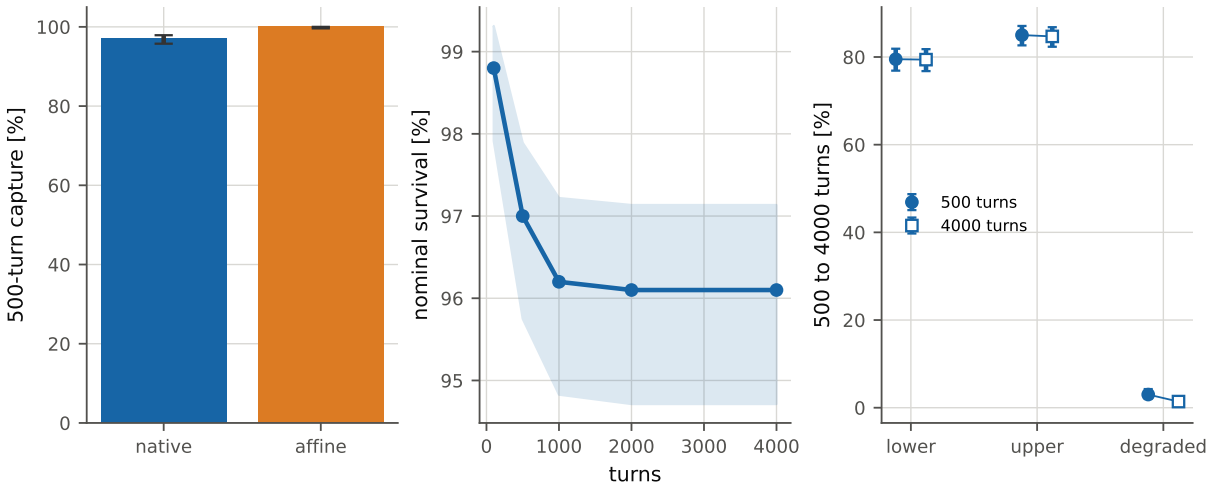


Figure 4: Final-point validation with the 3-D NKM, design beam and reference chamber. Left: native and affine-element backends at 500 turns on the same two 500-particle bunches (seeds 1001/1002; 95 % particle Wilson bars). Centre: native nominal survival with turn count (same particles; Wilson band). Right: 500-to-4000-turn survival for interpolated lower/upper angular-edge candidates and corrected ring 5501, each with two fresh 500-particle bunches (seeds 2001/2002); circles/squares denote 500/4000 turns, with 95 % Wilson bars. Intervals are conditional on the selected models and ring, not machine-population uncertainty.

fresh bunches gave 93.0 % (91.7–94.0 %) for that beam and septum-only aperture. This independent sample reduces reuse of selection noise; it does not remove dependence on the searched grid or model. The design-beam/chamber studies retain these knobs without reselection and therefore test a conditional operating point rather than a new global optimum.

We estimate the connected operating window above 80 % capture around that fixed point. Linear interpolation gives angle offsets from -0.52 to 0.24 mrad (width 0.76 mrad), and position offsets span -0.67 to 1.32 mm (width 1.99 mm). Each value uses seeds 1001/1002 and 500 particles per seed. A fixed 0.2 mm/mrad grid is extended if a boundary remains above threshold, with a maximum offset of 3; crossing cells are refined once to 0.1 spacing and linearly interpolated. Grid resolution is not a confidence interval for a crossing. The long-turn edge checks use fresh particles and do not redefine the selected 500-turn window.

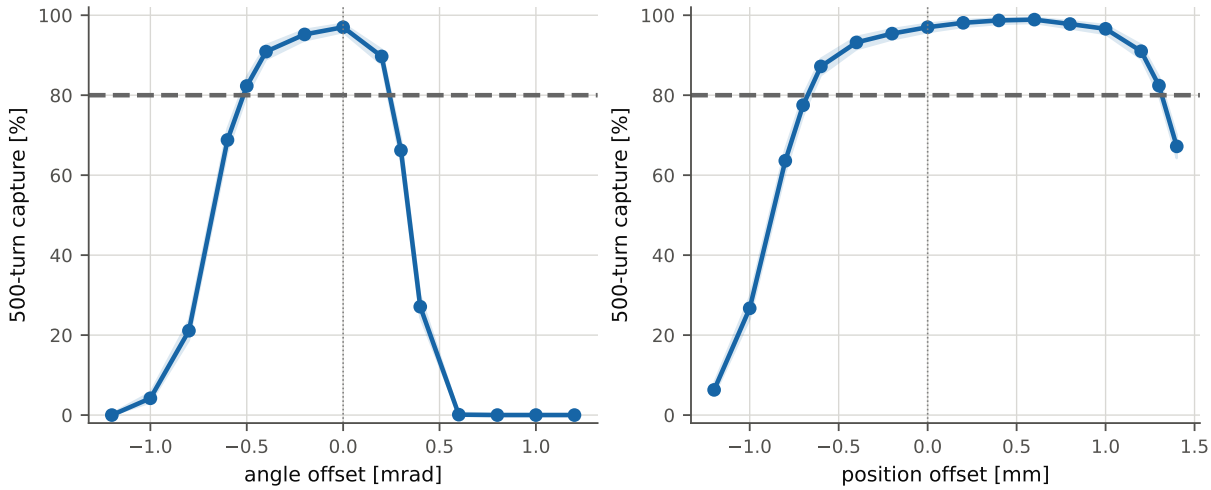


Figure 5: Design-beam operating window with the reference chamber and fixed NKM settings. Left/right: capture versus septum-exit angle/position offset, 500 native turns, two 500-particle bunches (seeds 1001/1002) per point. Bands are conditional 95 % particle Wilson intervals; the dashed line marks 80 %. Crossing cells have 0.1 mrad/mm spacing after refinement.

5 Survival, acceptance and sensitivity

Tracking 1000 particles for 4000 turns (seeds 1001, 1002), about one damping time, 93.0 % survive 500 turns, 89.6 % survive 2000 turns and 89.5 % the full run (95 % particle Wilson interval 87.4–91.3 %, conditional on these two bunches); 3.5 % of the particles are lost after turn 500 (figure 6, right). A single 2000-particle bunch gives 92.4 % (91.2–93.5 %): 17 particles are lost at the septum on later turns, 135 elsewhere in the ring, 0 at creation and 0 inside the magnet. With a uniform rectangular chamber of half-height 5 mm and half-width 15, 12, 11 or 10 mm at every element, capture is 86.3, 83.8, 76.0 and 47.9 %, respectively. The loss at 15 mm is vertical: a 10 mm half-height restores 92.4 %, and an open horizontal aperture with a 5 mm half-height gives 86.3 %. On the same particles the 5 mm half-height costs the same with and without the NKM vertical kick (93.0 to 86.3 % and 94.8 to 88.0 %), which is consistent with horizontal-to-vertical transfer in the ring; no unique coupling resonance is diagnosed here. At the selected point the NKM vertical kick itself costs only the difference between 94.8 and 93.0 %, less than at the earlier point of section 3, because the centroid now sits near the kick extremum, where the linear vertical focusing term $-\partial\Delta x'/\partial x$ vanishes. The vertical stay-clear at large horizontal amplitude is therefore a design constraint of the injection straight. Random relative gradient errors of 0.05 % and 0.10 % on all quadrupoles and 0.1 % on all sextupoles (10 seeds each, no misalignments, no correction) reduce the negative DA edge to between -11.0 and -13.0 mm and

between -9.5 and -12.0 mm, and give mean capture 85.6 % (range 77.6–89.6 %) and 69.9 % (58.0–80.0 %). Because these errors are uncorrected, they define selected sensitivity scenarios rather than predict a corrected machine; the injection DA of the corrected KLS lattice is the decisive input for an operational estimate.

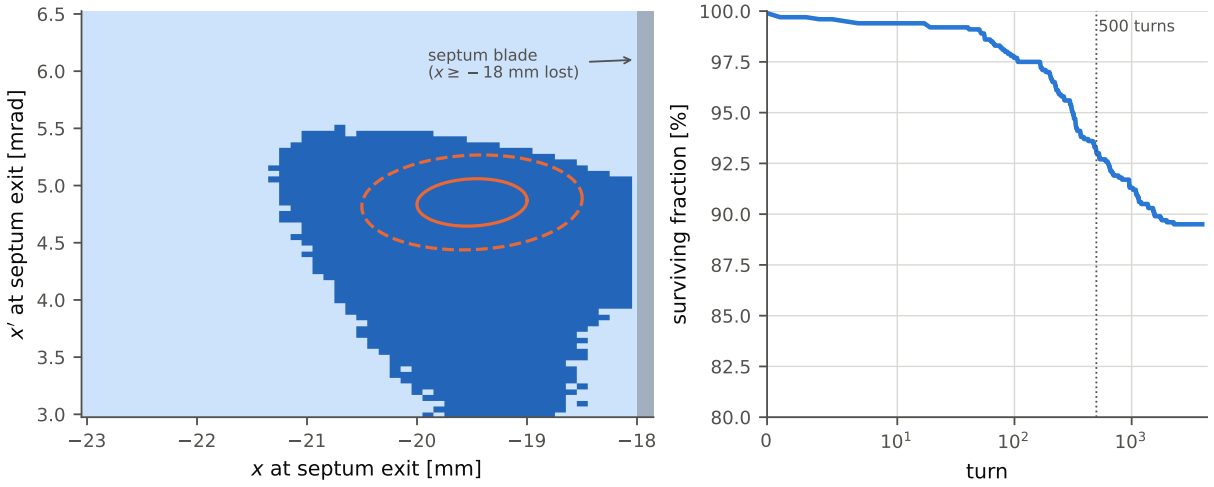


Figure 6: Left: single-particle survival for 500 turns in septum phase space ($\delta = 0$, 3692 particles) with the injected-beam 1σ (solid) and 2σ (dashed) ellipses and the septum blade (grey). Right: surviving fraction of the injected bunch versus turn at the selected point.

A separate selected corrected-ring scenario applies correlated 30 μm girder offsets, 30 μm independent magnet offsets, 100 μrad rolls, gradient/sextupole errors and skew-quadrupole components, all explicit sensitivity assumptions. Ten seeds (5501–5510) use the 120 HVC correctors and BPM markers with a truncated-SVD response, followed by bounded Q11/Q12 tune correction. The selected rms criterion is 100 μm; an earlier 20 μm target was not attained. Maximum achieved BPM rms residuals are 84.6/56.2 μm, and the largest fractional-tune residual is 9.1×10^{-6}. The full DA grids (figure 7) define the subsequent chamber/error scenario; they are not a commissioned-machine prediction.

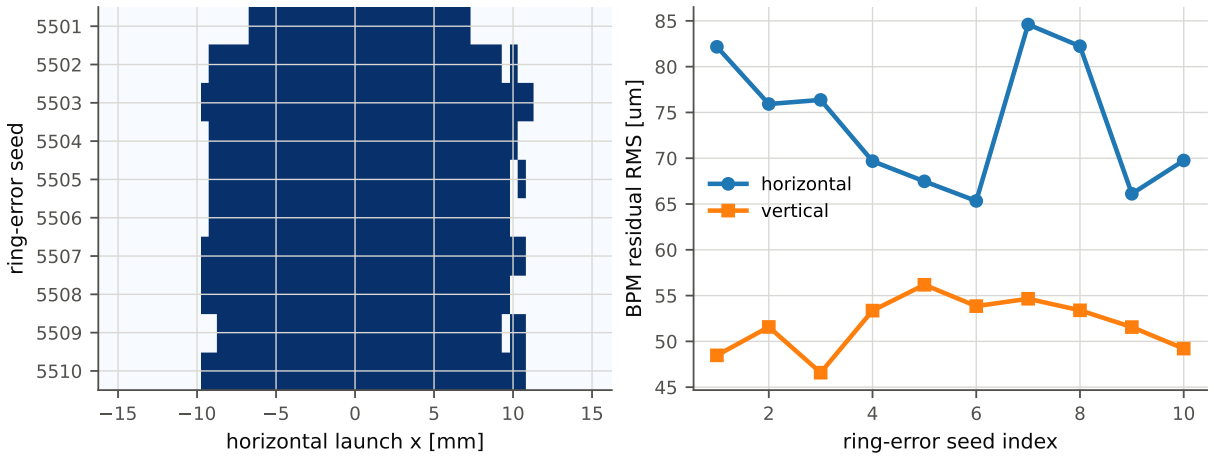


Figure 7: Selected corrected-ring scenario, ten fixed seeds. Left: deterministic 500-turn native survival on a 0.5 mm horizontal launch grid, with RF/radiation and the reference chamber. Blue/white cells denote surviving/lost launches. First failed grid points are not exact stable edges. Right: horizontal and vertical BPM rms residuals after SVD orbit and bounded tune correction. Curves show seed variation, without a particle-sampling interval.

The chamber reference uses arc half-width/height 12/9 mm, centre-bend half-height 4.87 mm,

and an inscribed ellipse where the octagonal corners are unspecified. An assumed rectangular NKM straight has half-width/height 30/15 mm. Nineteen identified ordinary straights receive 3 m IVU segments; their assignment is a scenario, not a surveyed map. The reference open half-gap is 9 mm; a 2.5 mm closed magnetic half-gap is a stress case and does not account for a liner. The separate incoming transport retains its source lattice and septum creation cut; its chamber has not been surveyed for this alternative injection scheme.

For the 7.67 nm rad design booster beam, the reference chamber gives 97.0 % capture at 500 turns (95 % particle Wilson interval 95.7–97.9 %; two 500-particle bunches, seeds 1001/1002). The paired dipole gives 97.3 %; the NKM-minus-dipole difference is -0.3 percentage points (95 % paired-particle bootstrap interval -0.9 to +0.3; 10 000 draws, seed 1729). Arc half-widths 10, 12 and 15 mm give 81.7, 97.0 and 99.7 % at each tested half-height (4.87, 7.5, 9 mm). Thus 10 mm is the smallest tested width exceeding 80 % in this ideal-ring sample, while 12 mm retains the reference result; neither is a continuous minimum or a machine tolerance. Across the ten corrected rings, mean capture is 61.7 %, median 63.7 %, tenth percentile 52.3 %, and range 2.7–87.4 %. These are different descriptors of a finite scenario population.

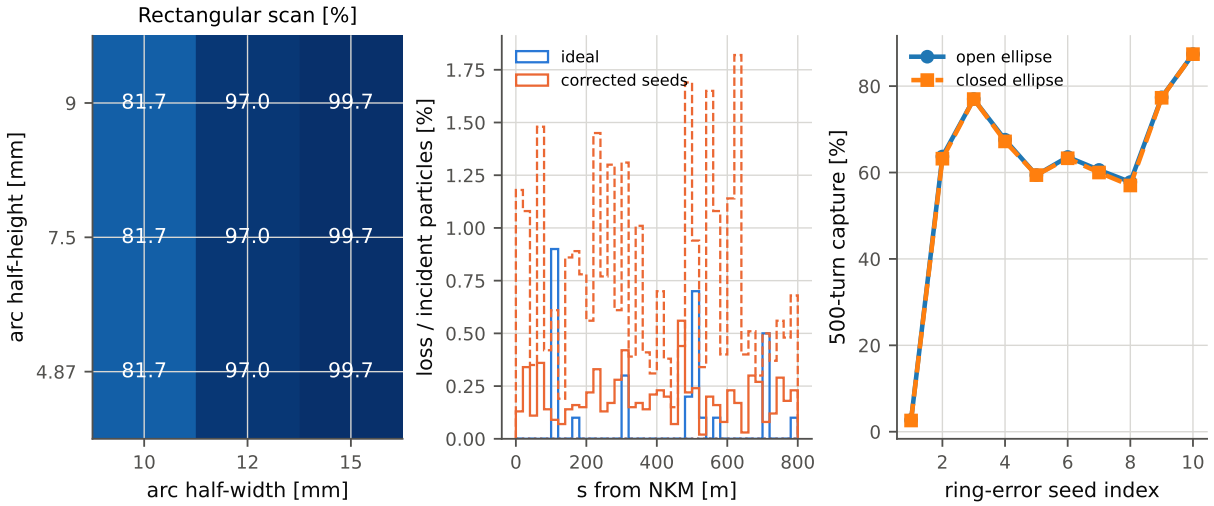


Figure 8: Declared chamber with the design booster beam, 500 native turns and two seeds of 500 particles per ring/profile/model. Left: ideal-ring capture versus arc half-apertures, with other region limits retained. Centre: ring losses per incident particle in 20 m bins from the NKM centre; solid/dashed/dotted denote horizontal/vertical/dynamical losses. For elliptic apertures the larger normalized coordinate defines the plane. Right: capture for each corrected ring with open and closed IVUs. Points are conditional sample fractions; the loss histograms and finite ring-seed variation have no assigned confidence band.

6 Stored-beam transparency

The kick of the four-loop magnet is odd in x on the midplane and its near-axis expansion has a leading cubic term, $\Delta x' \simeq c_3 x^3$ with $c_3 = 2.4 \times 10^4\,\mathrm{rad\,m^{-3}}$ at $S = 0.85$. For an aligned kicker and a centred, symmetric Gaussian beam the ideal centroid kick therefore vanishes exactly. This does not imply unchanged finite-beam moments: deterministic quadrature gives single-pass projected emittance changes of 4.5×10^{-7} horizontally and 1.6×10^{-7} vertically. Orders 20, 40 and 60 per axis agree at the reported precision. The meaningful quantities are the response to an offset between kicker and stored orbit and the emittance change, evaluated deterministically from the exact field by Gauss–Hermite quadrature (figure 9, right). Offsets of 0.1, 0.2 and 0.5 mm give centroid amplitudes A_x of 0.016, 0.10 and 1.54 of the rms size, and the spread of the kick across the beam increases the emittance about the displaced centroid by 9.4×10^{-5}, 1.2×10^{-3}

and 4.2×10^{-2}; a 0.2 mm vertical offset gives 0.17 of the vertical rms size. Once the centroid oscillation has also filamented, the rms emittance grows by the factor $1 + A_x^2/2 + \Delta\varepsilon_x/\varepsilon_x$, i.e. 1.007 at 0.2 mm and 2.22 at 0.5 mm. Native tracking uses two seeds (11 and 12), each with 300 paired particles, with radiation on. The maximum centroid difference is measured at the ends of turns 1–50, and projected emittance at the end of turn 1000: $A_x/\sigma_x = 0.10$ and 1.47, and mean paired emittance ratios 1.01 and 2.24. These tracked observables need not equal a fully filamented analytic estimate. An offset of 0.2 mm thus keeps the stored-beam disturbance at about a tenth of its size, whereas 0.5 mm is not transparent. The bicubic interpolation of the map agrees with the exact field to 9×10^{-11} rad near the axis, whereas bilinear interpolation errs by 1×10^{-6} rad and overstates the 0.1 mm response nineteen-fold (0.29). The calibrated dipole control would displace the stored beam by about 2495 rms sizes in a linear estimate; this hypothetical figure only illustrates why a uniform kicker cannot be used for top-up.

The assumed vertical-to-horizontal emittance ratio is 0.10, as supplied by the author; for a fixed vertical dipole kick, $A_y/\sigma_y \propto \varepsilon_y^{-1/2}$. Individual wire displacements and independent loop-current errors break the cancellation (tables 2 and 3). The one-error-at-a-time wire limits range from 0.29 to 2.04 μm for both centroid amplitudes to remain below $0.1\,\sigma$. Independent loop currents require relative balance of about 1.1×10^{-4}. In series powering, a common current error preserves dipole cancellation provided the geometry remains symmetric; geometry errors still matter. These are local linear tolerances, not an allocation for simultaneous errors or a manufacturing specification.

Table 2: On-axis wire sensitivity. Wire numbers follow the four quadrant loops, inner then outer wire. u is the displaced coordinate. Dipole derivatives $D_{x,y} = \partial K_{x,y}/\partial u$ are in rad/m; $Q_n = \partial J_{xx}/\partial u$ and $Q_s = \partial J_{xy}/\partial u$ are in rad/m^2, with $J = \partial(K_x, K_y)/\partial(x, y)$, $J_{yy} = -J_{xx}$ and $J_{yx} = J_{xy}$ to numerical precision. t_x, t_y are single-error 0.1-sigma limits in μm. A dash indicates no resolved first-order dipole sensitivity; higher orders remain. Deterministic central differences, step 0.1 μm; no statistical interval.

Wire	u	D_x	D_y	Q_n	Q_s	t_x	t_y
1	x	-0.000	+0.357	-67.4	+67.4	–	0.29
1	y	+0.357	-0.000	+67.4	+67.4	0.54	–
2	x	+0.000	-0.096	+9.3	-9.3	–	1.10
2	y	-0.096	+0.000	-9.3	-9.3	2.04	–
3	x	-0.000	-0.357	+67.4	+67.4	–	0.29
3	y	-0.357	-0.000	+67.4	-67.4	0.54	–
4	x	+0.000	+0.096	-9.3	-9.3	–	1.10
4	y	+0.096	+0.000	-9.3	+9.3	2.04	–
5	x	-0.000	+0.357	+67.4	-67.4	–	0.29
5	y	+0.357	-0.000	-67.4	-67.4	0.54	–
6	x	+0.000	-0.096	-9.3	+9.3	–	1.10
6	y	-0.096	+0.000	+9.3	+9.3	2.04	–
7	x	-0.000	-0.357	-67.4	-67.4	–	0.29
7	y	-0.357	-0.000	-67.4	+67.4	0.54	–
8	x	+0.000	+0.096	+9.3	+9.3	–	1.10
8	y	+0.096	+0.000	+9.3	-9.3	2.04	–

The controls (figure 9, left) separate the cost of nonlinearity: the linearised and dipole kicks capture 97.8 and 97.0 %, the paired NKM-minus-dipole difference is -4.1 points (95 % paired-bunch bootstrap interval -4.7 to -3.2), conditional on this larger-emittance sample; with the kicker off no particle survives.

Table 3: Independent-loop relative-current sensitivity. Dipole derivatives are in rad per relative current; normal and skew quadrupole derivatives are in rad/m per relative current. The final column is the joint two-plane 0.1-sigma single-loop limit in units of 10^{-4}. Deterministic derivatives with relative-current step 10^{-5}; no statistical interval.

Loop	D_x	D_y	Q_n	Q_s	Limit
1	-0.0009	-0.0009	+0.000	-0.262	1.15
2	+0.0009	-0.0009	+0.000	+0.262	1.15
3	+0.0009	+0.0009	+0.000	-0.262	1.15
4	-0.0009	+0.0009	+0.000	+0.262	1.15

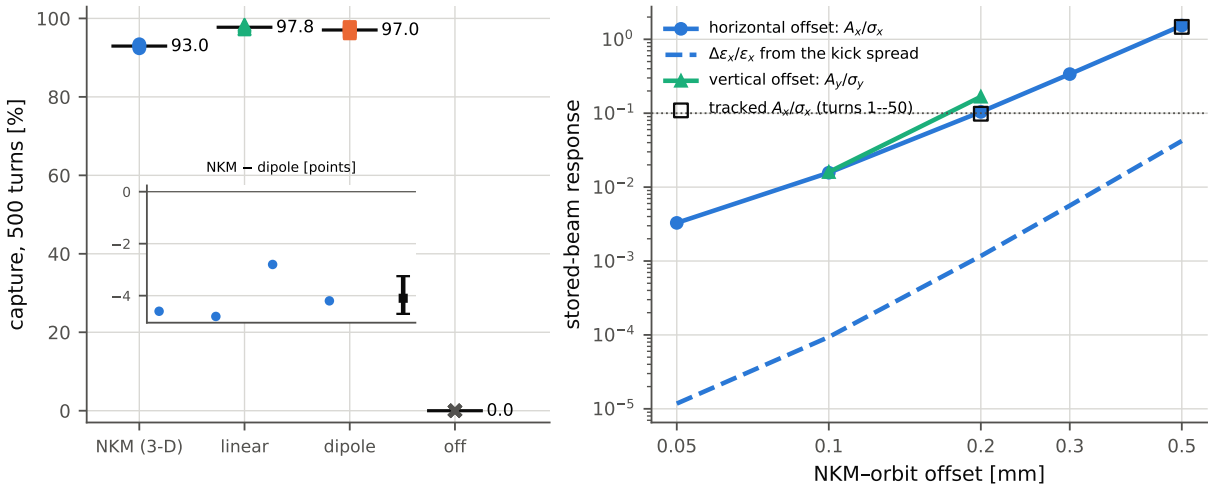


Figure 9: Left: 100/10 nm rad sensitivity case with the septum aperture and no additional chamber; four fresh 500-particle bunches at the selected point for the 3-D NKM, the calibrated linearised and dipole controls and the kicker off (markers: bunches; bars: mean). The inset shows per-bunch NKM-minus-dipole differences (dots), their mean (square) and a paired-bunch bootstrap 95 % interval (10 000 resamples, seed 1729; only four bunches). Right: deterministic centroid amplitude and filamented emittance growth versus NKM-to-orbit offset. Open squares are the mean of two 300-particle tracked seeds, using maximum centroid differences over ends of turns 1–50; they are not 1000-turn amplitudes. Emittance ratios in the text use the end of turn 1000. No sampling interval is assigned to deterministic quadrature.

7 Robustness and tolerances

7.1 Error budget

Table 4 defines the final untruncated Gaussian rms budget. Booster horizontal emittance 7.67 nm rad and energy spread 0.106 % are design values [24]; extraction Twiss, bunch length and the 0.10 vertical-emittance ratio are assumptions. No public extraction-orbit or septum-current stability was supplied, so the corresponding errors remain sensitivity parameters. Small exit Twiss mismatch is not used as evidence of capture robustness.

Table 4: Final rms error budget. S: static machine draw; J: shot draw. Unsourced entries are explicit sensitivity assumptions with the tested effect stated. Corrected-ring errors are a separate finite scenario population.

Term	RMS	Class	Source or reason
Booster centroid x/x'	0.5 mm / 0.2 mrad	J	Extraction stability unavailable; scan scales 0–1
Booster beta mismatch	5%	S	Assumed extraction-optics sensitivity
Emittance / mean energy	10% / 0.1%	J	Assumed fluctuations about design beam
Energy-spread variation	10^{-4} absolute	J	Assumed variation of within-bunch spread
BTS quadrupole gradient	0.1%	S	Assumed focusing error
BTS quadrupole x/y offset	100 μm each	S	Assumed alignment error
BTS quadrupole roll	0.5 mrad	S	Assumed coupling error
BTS dipole field	0.05%	S	Assumed rigidity/trajectory sensitivity
Septum position	0.1 mm	S	Assumed blade geometry error
NKM / residual orbit offset	0.2 mm each	S	Assumed relative alignment sensitivity
Common NKM amplitude	0.5%	J	Assumed pulser fluctuation, not a measured specification
Thin-septum strength	0.1% relative	J	Assumed; nominal bend 10 mrad [22]
RF phase	0.001 rad	J	Assumed injection-phase sensitivity
Arrival relative to NKM firing	2 ns	J	Assumed peak-phase pulse timing

The published bump-kicker reproducibility is not treated as an NKM or septum specification. At the lattice RF frequency, phase error shifts the injected longitudinal coordinate by $c\Delta\phi/(2\pi f_{\mathrm{RF}})$. Independent arrival relative to NKM firing scales the 5.3 μs half-sine field at peak phase and does not add another longitudinal displacement. The separate thin-septum error adds canonical momentum $(1+\delta)\,0.01\,\epsilon_{\mathrm{sept}}$ in radians. No equivalent transverse timing-kick term is added. Longitudinal quadrupole placement, supply quantization and a scalar ring-beta perturbation are not modelled and are not counted as applied errors.

7.2 Combined errors and tolerance curves

Three seeds (42, 123, 777) generate 150 joint machine/shot realizations, each with 200 Gaussian particles and 500 native turns. Each realization independently draws one of the ten corrected rings with replacement, plus the errors in table 4. The bootstrap unit is the joint realization, conditional on that finite ring population; it is not an estimate over all possible commissioning errors or repeated shots on one machine. Intervals use 10 000 percentile-bootstrap resamples, seed 1729.

Ideal static steering transports the zero-input reference through the perturbed BTS, giving (r_x, r_{p_x}), then translates the bunch at the septum by $(x, p_x) \mapsto (x - r_x, p_x - r_{p_x})$. The other coordinates are unchanged. This removes no shot jitter, vertical trajectory error or optics distortion, and does not model a fitted orbit-correction system.

The 3-D NKM mean is 14.8 % (95 % interval 11.2–18.8 %), median 1.0 %, tenth percentile 0.0 %, and fraction of realizations at or above 80 % is 2.7 %. The paired dipole mean is 22.5 % (18.3–27.0 %). The NKM-minus-dipole mean difference is -7.7 percentage points (-10.3 to -5.4). These controls compare models on identical particles and errors; they are not separately optimized injection systems.

The jitter scan retains the same 50 seed-42 realizations, particle seeds and ring identities while scaling only booster position and angle jitter. Table 5 reports differences from zero jitter. Quarter jitter corresponds to 0.125 mm and 0.05 mrad; its paired mean difference is +2.2 points, with interval -1.5 to +6.1. The interval does not establish the predeclared one-percentage-point loss margin; the candidate jitter target remains precision-limited. Position-only and angle-only reference jitter are evaluated separately. Mean, median, tenth percentile and fraction above threshold remain distinct; no 80 % design specification follows from a nominal ideal-ring value.

Table 5: Paired jitter sensitivity, 50 joint realizations per row. Capture summaries and the fraction of realizations at or above 80 % are percentages; differences from zero jitter and their paired 95 % bootstrap intervals are percentage points.

Scale	Mean	Median	p10	Fraction ≥ 80%	Difference	95% interval
0	34.1	28.2	0.0	4	+0.0	[+0.0, +0.0]
0.25	36.3	31.5	0.0	6	+2.2	[-1.5, +6.1]
0.5	29.4	24.0	0.0	4	-4.7	[-9.6, +0.0]
0.75	22.9	7.0	0.0	4	-11.2	[-17.8, -5.2]
1	19.8	3.5	0.0	4	-14.3	[-21.3, -7.8]
x_only	30.8	21.5	0.0	6	-3.3	[-8.5, +1.5]
xp_only	20.8	3.8	0.0	6	-13.2	[-19.6, -7.2]

Partial Spearman tests control the other sampled errors and ring-identity indicators. Their asymptotic two-sided t-test probabilities receive Holm adjustment across the sampled-error hypotheses. Scalar errors enter as magnitudes and vector errors as rms values. The smallest Holm-adjusted probabilities below 0.05 correspond to booster angle ($\rho = -0.51$, $p_{\mathrm{Holm}} = 2.20 \times 10^{-8}$). These describe conditional associations, not causal error rankings; the complete coefficient and adjusted-probability table is included in the numerical evidence.

For reduced capture-based tuning, each of the 50 seed-42 static machines receives 100 fresh training particles with shot fluctuations disabled. A single bounded coordinate sweep evaluates seven settings of $(x_{\mathrm{inj}}, x'_{\mathrm{inj}}, S)$: bounds $[-22, -18.2]$ mm, $[3.5, 6.5]$ mrad and $[0.7, 1.2]$, with steps 0.5 mm, 0.3 mrad and 0.05. Strict improvements are accepted and ties retain the current setting. The fixed seven-evaluation budget is the termination rule; settings remain feasible within those bounds. Independent held-out 200-particle samples then restore the joint shot errors at zero and quarter booster jitter. The tuned-minus-untuned gains are +11.6 points (+7.8 to +15.5) and +5.5 points (+1.1 to +10.0), respectively, with paired 95 % realization-bootstrap intervals. The corresponding tuned means are 45.7 and 41.8 %; fractions of realizations above 80 % are 12.0 and 10.0 %.

Training capture has one-percentage-point counting resolution. No BPM calibration, resolution, charge-monitor noise or actuator-error model is included, so this reduced ideal-observable search quantifies recovery without demonstrating operational beam-based feedback. Held-out results are never used to choose knobs, and the dipole control is not retuned.

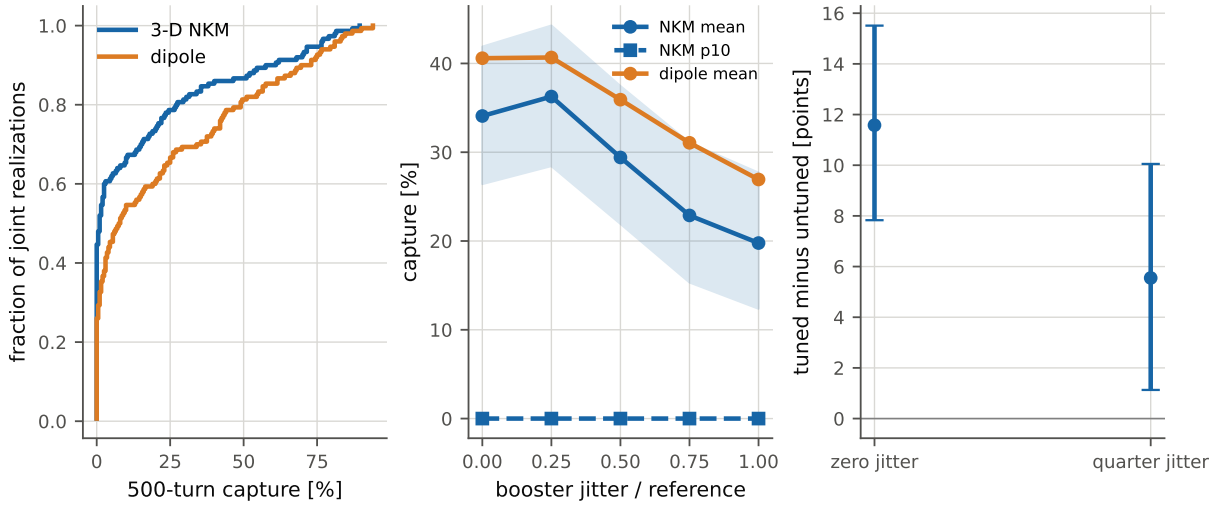


Figure 10: Final design-beam robustness with the reference chamber. Left: empirical cumulative capture distributions for 150 paired joint-error draws, conditional on ten corrected rings. Centre: NKM mean and tenth percentile and dipole mean versus jitter scale (50 paired draws); shading is a 95 % realization-bootstrap interval of the NKM mean. Right: held-out tuned-minus-untuned mean capture at zero and quarter jitter, with paired 95 % intervals (50 machines, 100 training and 200 held-out particles each). Bootstrap count 10 000, seed 1729. All tracking uses 500 native turns.

8 Pulsed-magnet considerations

The results above apply to an ideal single-pass spatial field. Native tracking of the nominal centroid observes positions -2.9, 6.2 and 9.0 mm after turns 1–3 at the NKM centre. The corresponding straight-line full-field kicks are 0.54, 3.71 and 4.87 mrad. Assigning one tenth of the conditional angular-window width (0.076 mrad) as a residual allowance is a design assumption; it gives first/second-return fractional limits 0.14/0.02.

For a half-sine of total duration τ, injection at its peak ($t = \tau/2$) gives subsequent field fraction $\max[0, \cos(\pi T_0/\tau)]$, with finite support. The native-centroid allowance gives $\tau \leq 5.85\,\mu$s; zero field on every return requires $\tau \leq 5.33\,\mu$s. A 5.3 μs peak-phase example meets the latter condition, whereas 8 μs retains 0.50 of the peak on the first return. Deterministic timing offsets obey $1 - \cos(\pi\Delta t/\tau)$, giving 7.7×10^{-8}–4.9×10^{-6} for 1–5 ns and 5–8 μs. The 2 ns Gaussian rms in the error model is a distribution, not a hard timing bound.

A train with its first bunch at the peak samples less field on later bunches: for the 5.3 μs example, 100, 200 and 500 ns train durations give approximately 0.18, 0.70 and 4.4 % variation. At an offset kicker the stored-beam centroid response scales with that field; aligned centroid cancellation does not imply zero finite-beam moment change. On-axis residual dipole and gradient allowances are 4×10^{-5} and 7×10^{-5} of the corresponding peak quantities. A hypothetical thin cylindrical conducting shell has diffusion time $\tau_{\mathrm{eddy}} \simeq \mu_0 \sigma a t/2$: at $a = 15$ mm, a 1 μm coating with $\sigma = 10^6$ S/m gives 9.4 ns, whereas a 1 mm wall with $\sigma = 5 \times 10^7$ S/m gives 0.47 ms. These are order-of-magnitude scenarios, not predictions for the unspecified NKM chamber. Coating-induced multipoles require chamber-specific analysis [11]; no time-dependent capture or operational pulse feasibility is claimed.

The sagitta ratio $\eta = |K|L/(8w)$ compares trajectory displacement with the horizontal kick-lobe full width at half maximum w. Figure 11 varies length, integrated strength and a second wire family scaled transversely by 1.25. It is a deterministic model comparison, not a universal capture criterion. A sliced vector map retains self-deflection and the physical vertical kick; the local focusing sign changes between flanks of the kick extremum. The original vertical-sign error

was an input error, not a failure inherent to kick maps. Capture controls throughout are paired model comparisons on identical particles, not independently optimized injection systems.

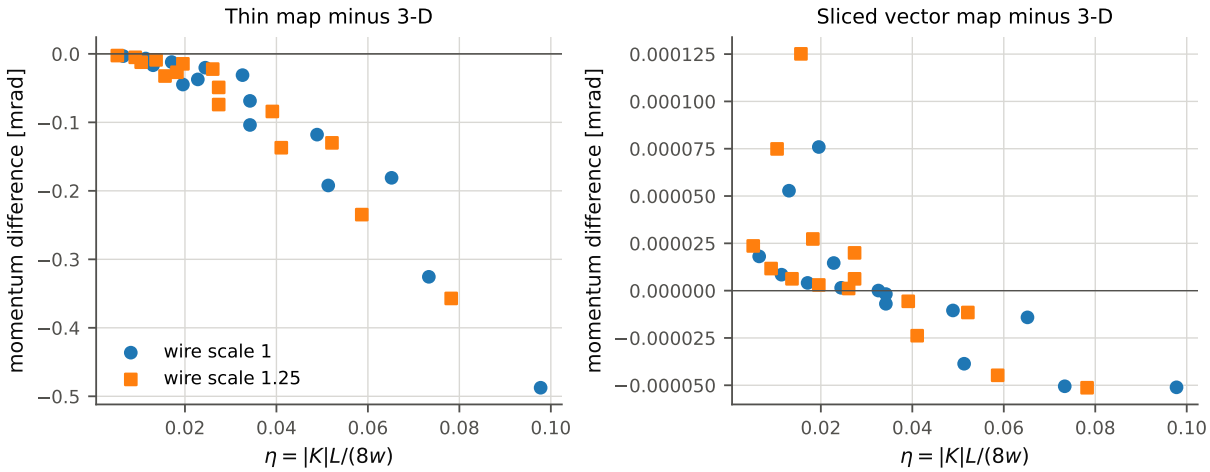


Figure 11: Single-centroid momentum differences versus η for five lengths (0.2–1.0 m), three integrated strengths and two transverse wire families. Left: thin vector map minus 3-D field. Right: vector map distributed over 1280 slices minus the 3-D field. Width w is the negative kick lobe's full width at half maximum. The centroids have nonzero vertical position and angle; these deterministic comparisons have no sampling interval.

9 Discussion and limitations

Four results generalise beyond this design. First, the kick map is not a controlled description of a strongly nonlinear injection kicker: the self-deflection inside the magnet, with sagitta $|K|L/8$, shifts the matched angle and changes the error sensitivity in a sign-dependent way, and the magnet length is a design variable that trades nominal amplitude against tolerance. Second, the curl-free vertical kick defocuses the injected bunch wherever the horizontal kick gradient is non-zero and costs up to several points of capture, which favours placing the centroid at the extremum; tabulated maps should be checked for $\partial\Delta x'/\partial y = \partial\Delta y'/\partial x$. Third, the transparency of an aligned NKM is a symmetry property; the engineering quantities are the kicker-to-orbit alignment, the on-axis residual field and the pulse length. Fourth, with an angular window of order 0.74 mrad and a position window of about 1.44 mm, the injected orbit at the septum, not the kick strength or the transfer-line optics, decides whether the window can be used: static transfer-line errors must be corrected, the booster jitter kept to a quarter of the assumed amplitude, and the remaining static offsets of kicker, orbit and septum budgeted together.

The limitations are explicit. The ring is an ideal lattice; ring errors and apertures enter only as selected sensitivity scenarios. The filament model contains no iron, chamber or eddy-current fields and no pulse time dependence. The error budget is assumed. Capture is survival over 500 turns, checked to one damping time at the selected point. As at BESSY II and SIRIUS [9, 15], sextupole optimisation for the injection DA is expected to enlarge the window and was not attempted.

10 Conclusions

For a 0.525 m wire NKM in the 4 GeV KLS design lattice, exact 3-D field tracking with a coupled transfer line gives 93.0 % capture at the selected point (97.0 % for an equally strong uniform kick), an angular window of 0.74 mrad at 80 % capture, and an ideal centroid kick that vanishes

for the aligned kicker and is 0.10 of the rms size at a 0.2 mm offset. A single thin map neglects self-deflection; vector maps can retain the vertical kick, whose original sign error is an input-convention error. Both corrections change capture by several points. Under the assumed errors the mean capture is 44.4 % with ideal static orbit correction and 40.5 % without; the quarter-jitter result is precision-limited relative to zero jitter, and a mean capture of 80 % additionally requires tighter static tolerances or a wider window.

Data, software and code availability

The simulation code, configurations and analysis scripts are available in the nkm-injection repository (`https://github.com/cspark7701/nkm-injection`). The figures and numbers are generated from the campaign artifacts listed, with SHA-256 digests, in `figure_provenance.csv` of the manuscript directory `docs/paper/generated`. A frozen evidence bundle with the selected simulation outputs, input files, code snapshot and rebuild instructions is provided to referees; an archived release with a persistent identifier will be created on acceptance.

Acknowledgments

The author thanks the accelerator physics group at Korea University. Generative AI assistance (Anthropic Claude, used through Claude Code) was used to implement simulation drivers and the field model, run the campaigns, audit references and draft parts of the manuscript; all results were checked and the author takes full responsibility for the content.

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